

# Network Security Operations Manual

## 1. Introduction to Network Security

Network security encompasses the policies, procedures, and technologies used to protect the integrity, confidentiality, and accessibility of computer networks and data. As organisations grow increasingly reliant on digital infrastructure, robust security practices have become essential to protect against evolving cyber threats.

This manual provides system administrators and security professionals with a comprehensive guide to designing, implementing, and maintaining a secure network environment. Topics range from fundamental firewall configuration to advanced intrusion detection strategies.

## 2. Firewall Configuration

A firewall is the first line of defence in any network security architecture. Modern next-generation firewalls (NGFWs) perform deep packet inspection (DPI), application-layer filtering, and stateful connection tracking. When configuring a firewall, administrators should adhere to the principle of least privilege: deny all traffic by default and allow only explicitly required communication paths.

Rule ordering is critical. Firewalls evaluate rules top-to-bottom and stop at the first match. Specific allow rules should precede broad deny rules. Regularly audit and prune stale rules to reduce the attack surface. Logging should be enabled on all deny rules to capture reconnaissance attempts and anomalous traffic patterns.

Zone-based segmentation separates the network into logical regions — DMZ, internal LAN, and management plane — each governed by distinct policy sets. Traffic crossing zone boundaries must be explicitly permitted, minimising lateral movement in the event of a breach.

## 3. Intrusion Detection and Prevention Systems

Intrusion Detection Systems (IDS) monitor network traffic for signatures of known attacks and anomalous behaviour patterns. An Intrusion Prevention System (IPS) extends this capability by actively blocking identified threats in real time. Both systems rely on signature databases that must be updated continuously to address newly discovered vulnerabilities.

Anomaly-based detection establishes a baseline of normal network behaviour and alerts on deviations. While more effective against zero-day attacks than signature-only approaches, anomaly engines require careful tuning to keep false-positive rates manageable. Machine learning models trained on labelled traffic datasets are increasingly used to improve accuracy.

Placement of sensors matters. Inline IPS sensors between the firewall and core switches inspect north-south traffic, while lateral sensor deployment captures east-west flows between internal segments — a common vector for ransomware propagation.

## **4. VPN and Encrypted Tunnels**

Virtual Private Networks (VPNs) extend a private network across a public medium by encapsulating traffic in encrypted tunnels. IPsec and TLS/SSL are the dominant protocols. IPsec operates at the network layer and is suited for site-to-site connectivity, while TLS-based solutions such as OpenVPN and WireGuard are preferred for remote-access scenarios.

WireGuard has gained rapid adoption due to its minimal codebase (~4,000 lines versus OpenVPN's ~70,000), which reduces the audit surface and improves performance. Its use of modern cryptographic primitives — Curve25519, ChaCha20-Poly1305, BLAKE2 — provides strong security with lower computational overhead than legacy IPsec cipher suites.

Always enforce certificate-based mutual authentication rather than pre-shared keys for production deployments. Rotate certificates annually and revoke compromised credentials immediately via a maintained Certificate Revocation List (CRL) or OCSP responder.

## **5. Incident Response Procedures**

A well-defined Incident Response Plan (IRP) reduces the mean time to detect (MTTD) and mean time to respond (MTTR) to security events. The six phases of the NIST incident response lifecycle are: Preparation, Detection and Analysis, Containment, Eradication, Recovery, and Post-Incident Activity.

During containment, isolate affected systems by removing them from the network while preserving forensic evidence. Memory dumps and disk images should be captured before remediation to support root-cause analysis. Chain-of-custody documentation is mandatory if legal proceedings are anticipated.

Post-incident reviews must produce actionable findings. Update detection rules, patch exploited vulnerabilities, and revise the IRP based on lessons learned. Tabletop exercises conducted quarterly help teams internalise procedures before a real incident occurs.

